

B.TECH CSE (AI/ML) · SESSION 2025-26

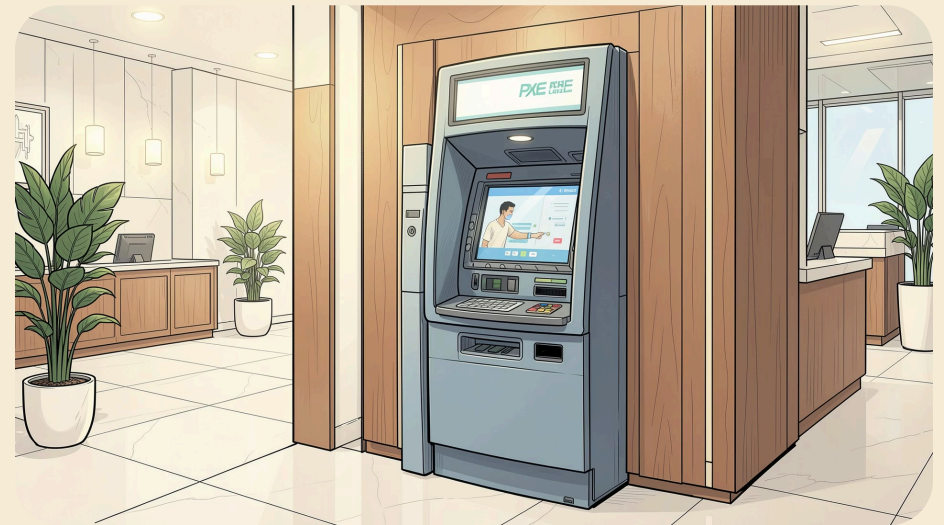
ATM Simulation Using Object-Oriented Programming (Java)

Student: [Aditya kumar] · ERP: [RU-25-10090]

Branch: B.Tech CSE (AI/ML) With IBM

Submitted To: [Mr. Santanu sasmal]

Institution: Rungta International Skills University



Introduction

An Automated Teller Machine (ATM) enables customers to perform banking transactions — cash withdrawal, balance inquiry, and more — without visiting a branch. This project simulates core ATM operations using Java and OOP principles.

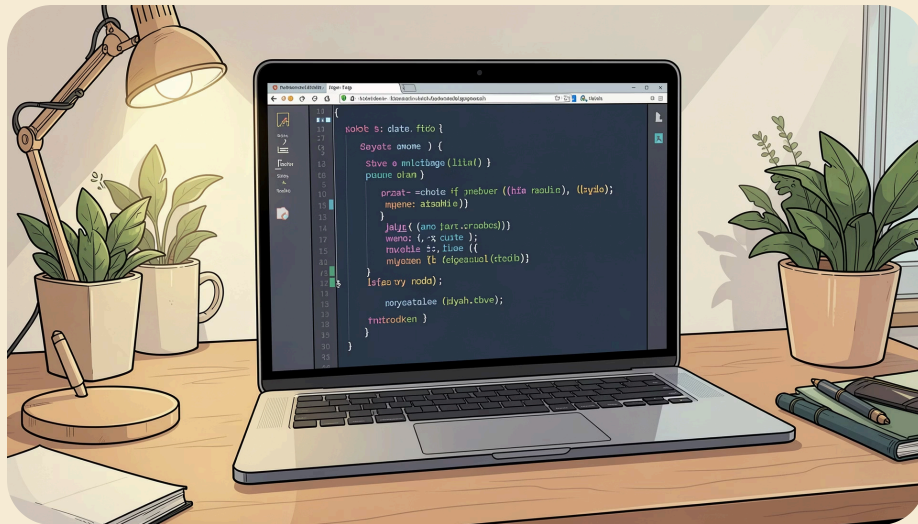


Project Objectives

- Secure PIN Authentication
- Cash Withdrawal Operations
- Real-Time Balance Inquiry
- Robust Error Handling

Technologies & OOP Concepts Used

Core Java & OOP Principles



Java Language

Core logic and program structure



Classes & Objects

Modular design with ATM and Account classes



Encapsulation

Data hiding for account security



Exception Handling

Graceful error management



Loops & Conditionals

Menu navigation and flow control



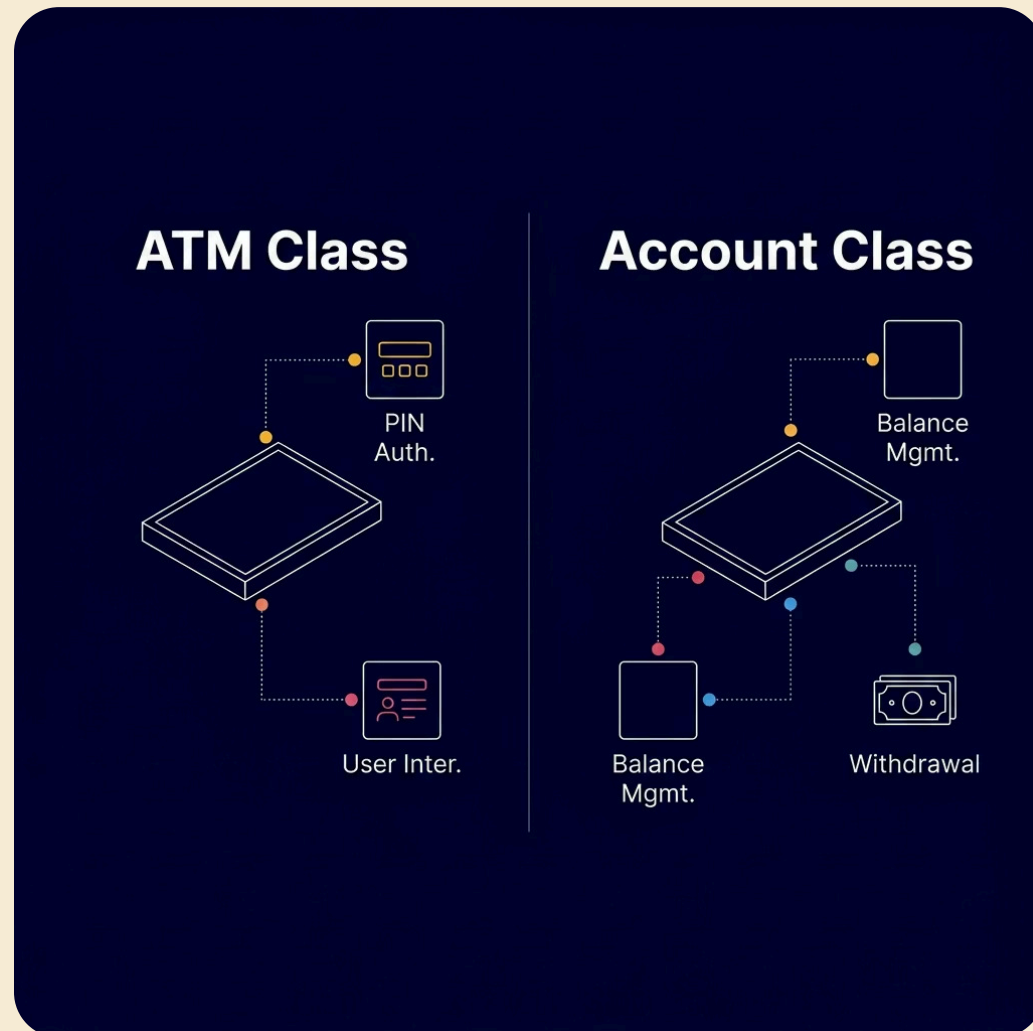
Static Variables

Shared state across instances

Development Environment: VS Code / IntelliJ IDEA

System Architecture

The project is structured around two core classes that collaborate to simulate a complete ATM experience — separating authentication and interaction logic from account data management.



ATM Class

- PIN Authentication module
- Menu-driven user interface
- Input validation and navigation

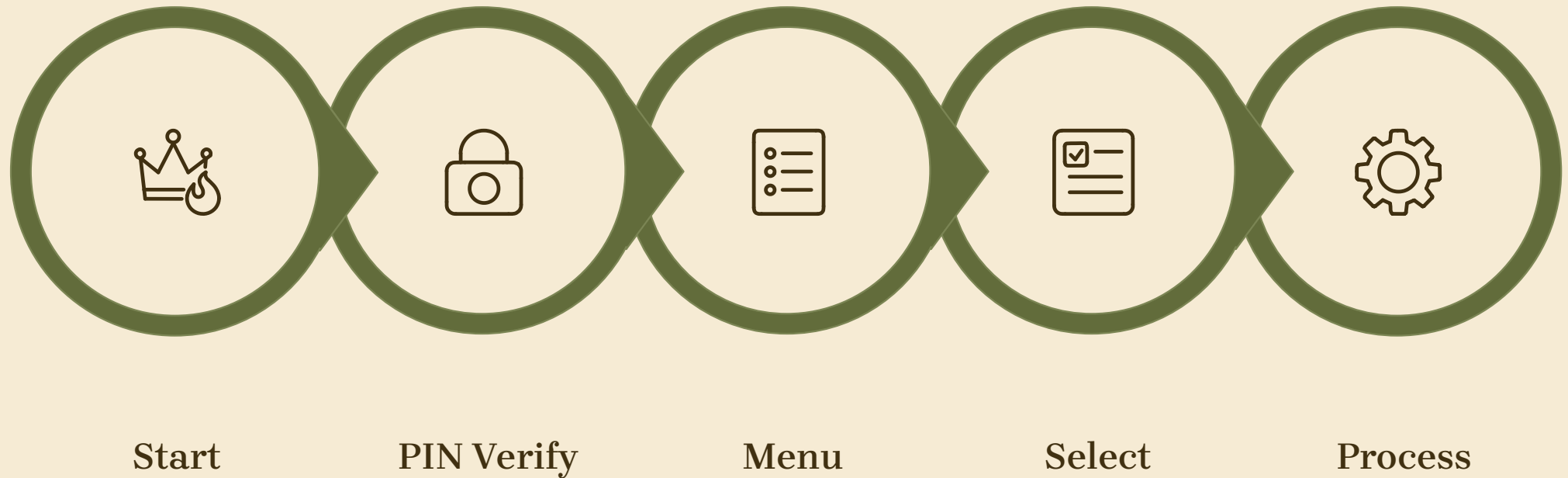
Account Class

- Encapsulated balance field
- Withdrawal method with validation
- Balance inquiry getter

i Encapsulation ensures account data is accessed only through controlled methods, preventing direct manipulation.

Working Flow

The ATM simulation follows a clear, sequential flow — from user authentication through transaction processing to session exit. Each step includes validation to ensure secure and reliable operations.



If an incorrect PIN is entered three times, the system locks access — mirroring real-world ATM security behavior.

Features & Exception Handling

Core Features

Secure PIN Authentication

3-attempt limit with
lockout

Cash Withdrawal

Validates amount and
balance

Balance Inquiry

Real-time account
balance display

Menu-Driven Interface

Intuitive console
navigation

Exception Handling

Invalid PIN

Alerts user after each
failed attempt

Invalid Input

Catches non-numeric
menu selections

Negative Amount

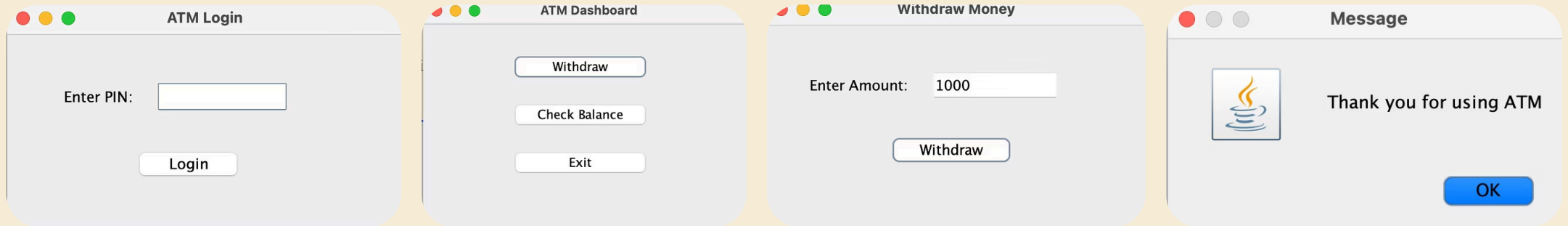
Rejects withdrawal
values below zero

Insufficient Balance

Notifies when funds are
inadequate

Program Screenshots

Key screens from the ATM simulation demonstrating the user journey from authentication through transactions and error handling.



Future Scope

The current console-based simulation provides a solid foundation. The following enhancements are planned to evolve this into a full-featured banking application.



Deposit & PIN Change

Add cash deposit functionality and allow users to update their PIN securely



Transaction History

Maintain a log of all past transactions for user reference and auditing



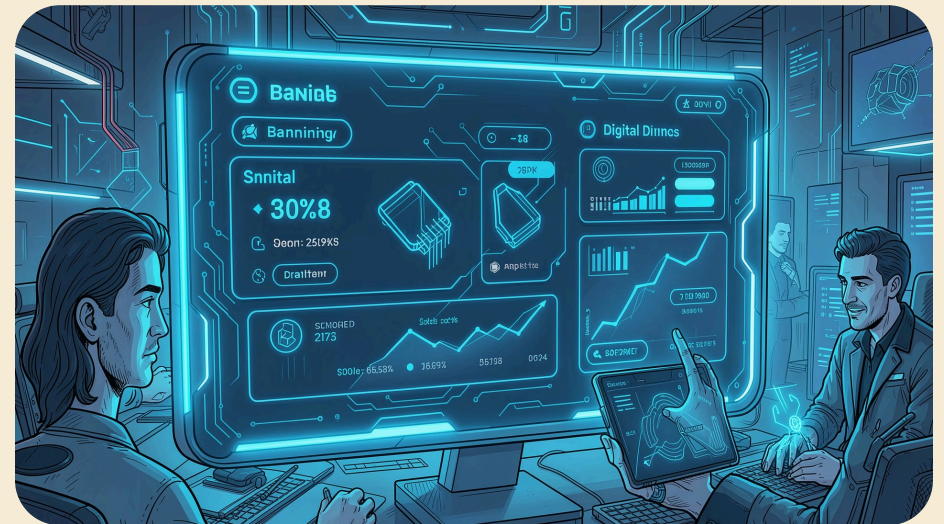
Database Integration

Connect to MySQL for persistent storage of account data across sessions



GUI & Mobile Extension

Migrate to Java Swing for a graphical interface, with potential mobile banking support



Conclusion

This project successfully demonstrates how core OOP principles in Java can model real-world banking systems with security, structure, and reliability.

✓ OOP Applied

Encapsulation, classes, and objects used effectively throughout

✓ Secure by Design

PIN authentication and input validation protect user data

✓ Robust Error Handling

Exceptions managed gracefully for a smooth user experience

✓ Real-World Simulation

Accurately mirrors core operations of a physical ATM

🕒 The project meets all stated objectives and serves as a strong foundation for advanced banking application development.

Thank You

We welcome your questions and feedback.

Project

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